

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458246.683 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.042
Transit depth (Rp/Rs)^2	3.78 +/- 1.62 [%]
Orbital Inclination (inc)	78.38 +/- 2.2
Airmass coefficient 1 (a1)	0.934 +/- 0.013
Airmass coefficient 2 (a2)	0.0353 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#1 - [441, 295]
Optimal Aperture	4.27
Optimal Annulus	4.27
Transit Duration (day)	0.034 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.042 vs 0.1594 (deviation 0.82σ)
Duration fit vs published	0.034 d vs 0.0512 d (deviation 1.08σ)
Mid-time O–C	-3.7 min
Depth SNR	4.6
Residual scatter	1.36 %

Light curve

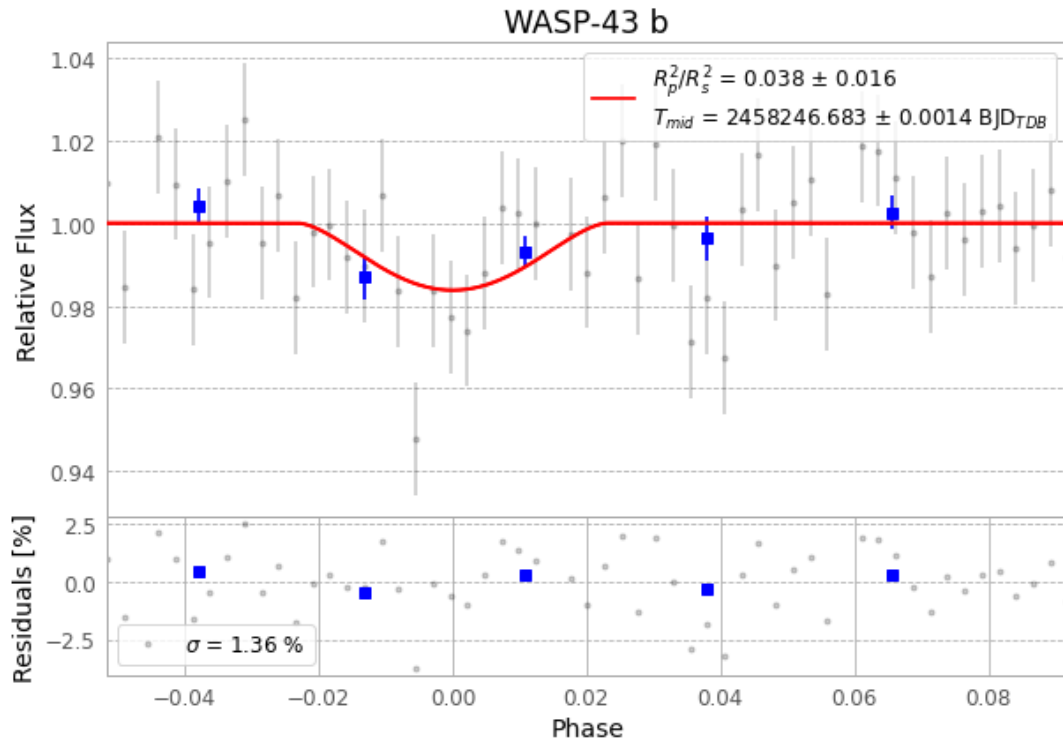


Figure 1 — FinalLightCurve_WASP-43 b_2018-05-07.png (SHA-256 81f1c2594d96d26d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 00be636caad063b8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

57 FITS frames (37 MB), WASP-43180508031812.FITS → WASP-43180508060612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180508.log (d2bb8ae733ad...), FinalLightCurve_WASP-43 b_2018-05-07.png (81f1c2594d96...), FinalLightCurve_WASP-43 b_2018-05-07.csv (d422f8e4e83e...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 14:04Z.